

Module 10: Epigenetics — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Explain why gene regulation is essential — not all genes are active in all cells
2. Describe the levels at which gene expression can be regulated
3. Explain how transcription factors control which genes are turned on or off
4. Describe the lac operon as a model of prokaryotic gene regulation
5. Define epigenetics and explain how cells with identical DNA can have different functions
6. Describe DNA methylation and histone modification as epigenetic mechanisms
7. Explain how environmental factors can influence gene expression without changing DNA sequence

The Big Idea: Same DNA, Different Cells

Every cell in your body contains the **same DNA**. A skin cell and a brain cell are genetically identical, yet they look and function completely differently. **Gene regulation** determines which genes are expressed (turned on) in each cell type. **Epigenetics** is the study of how expression patterns are established and maintained without changing the DNA sequence.

Key Terms to Know

Gene Expression Regulation

- **Gene Expression** — The process by which information in a gene is used to make a functional product (usually a protein)
- **Gene Regulation** — Mechanisms that control when, where, and how much of a gene product is made

- **Transcription Factor** — A protein that binds to DNA and promotes or inhibits transcription of a specific gene
- **Activator** — A transcription factor that increases gene expression
- **Repressor** — A protein that blocks transcription of a gene
- **Promoter** — DNA sequence where RNA polymerase binds to start transcription
- **Enhancer** — A distant DNA sequence that, when bound by activators, increases transcription (can be thousands of base pairs from the gene)
- **Silencer** — A DNA sequence that, when bound by repressors, decreases transcription
- **Operon** — A cluster of genes under the control of a single promoter (prokaryotes)
- **Housekeeping Gene** — A gene expressed in all cells at all times (e.g., genes for ribosomes, metabolic enzymes)

Epigenetic Mechanisms

- **Epigenetics** — Heritable changes in gene expression that do NOT involve changes to the DNA sequence itself
- **DNA Methylation** — Addition of methyl groups ($-\text{CH}_3$) to cytosine bases → typically silences gene expression
- **Histone** — Protein around which DNA is wrapped; modifications affect how tightly DNA is packaged
- **Histone Acetylation** — Adding acetyl groups to histones → loosens chromatin → gene expression **ON**
- **Histone Deacetylation** — Removing acetyl groups → tightens chromatin → gene expression **OFF**
- **Euchromatin** — Loosely packed chromatin; genes are accessible and can be transcribed
- **Heterochromatin** — Tightly packed chromatin; genes are inaccessible and silenced
- **Chromatin Remodeling** — Changes in chromatin structure that affect gene accessibility
- **X-Inactivation** — One X chromosome in female mammals is condensed into a Barr body (epigenetically silenced)

Post-Transcriptional Regulation

- **mRNA Stability** — How long an mRNA molecule lasts before being degraded; longer-lived mRNA = more protein
- **miRNA (MicroRNA)** — Small RNA molecules that bind to mRNA and block translation or trigger degradation
- **Alternative Splicing** — Different combinations of exons produce different mRNA variants (and different proteins) from the same gene
- **Post-Translational Modification** — Chemical changes to proteins after translation (folding, phosphorylation, ubiquitination) that affect their activity, location, or lifespan

Levels of Gene Regulation

Gene expression can be controlled at multiple levels:

Level	Mechanism	Example
1. Chromatin	DNA methylation, histone modification	Heterochromatin silences genes
2. Transcription	Transcription factors, enhancers, silencers	Activators turn on muscle-specific genes
3. Post-Transcription	mRNA splicing, mRNA stability, miRNA	Alternative splicing → different protein variants
4. Translation	Regulatory proteins, miRNA blocking ribosome	miRNA blocks translation of target mRNA
5. Post-Translation	Protein modification, degradation	Ubiquitin tags proteins for destruction

Transcriptional regulation (Level 2) is the most common and most energy-efficient control point — it's better to prevent an unnecessary mRNA from being made than to make it and then destroy it.

Model System: The Lac Operon (Prokaryotes)

The **lac operon** in *E. coli* is a classic example of gene regulation:

- **Without lactose:** The repressor protein binds to the operator and **blocks transcription** of lactose-digesting enzymes
- **With lactose:** Lactose (allolactose) binds the repressor, causing it to release from the operator → RNA polymerase can transcribe the genes → enzymes are made
- This is an example of **inducible regulation** — the genes are only expressed when needed

Why it matters: This demonstrates a fundamental principle — cells conserve energy by only making proteins when they are needed.

Epigenetics: Beyond the DNA Sequence

How Epigenetics Works

1. **DNA Methylation** — Methyl groups added to C bases (especially CpG sites) → gene silenced
2. Patterns can be inherited through cell division
3. Example: Genomic imprinting — some genes are expressed only from the maternal or paternal copy
4. **Histone Modification** — Chemical tags on histone tails determine chromatin accessibility
5. Acetylation → open → expression ON
6. Methylation (of histones) → can be activating or silencing depending on location
7. **X-Inactivation** — In female mammals, one X chromosome is randomly inactivated in each cell
8. Example: Calico cats — patches of orange and black fur reflect different X chromosomes being active in different skin cell lineages

Environmental Influences on Gene Expression

- **Nutrition:** Folic acid (a methyl donor) during pregnancy affects methylation patterns in offspring
- **Stress:** Chronic stress can alter methylation patterns that affect stress-response genes
- **Toxins:** Environmental chemicals can disrupt normal epigenetic programming
- **Exercise:** Physical activity has been shown to alter methylation patterns in muscle and fat cells

Key takeaway: Your genes are not your destiny. While you can't change your DNA sequence, environmental factors and lifestyle choices can influence which genes are active through epigenetic mechanisms.

Why This Matters

- **Cell differentiation** is controlled by gene regulation — stem cells become specialized by turning on and off specific gene sets
- **Cancer** can result from abnormal gene regulation (e.g., silencing tumor suppressors via methylation)
- **Personalized medicine** may target epigenetic modifications to treat disease
- **Identical twins** can develop different traits over time due to diverging epigenetic patterns

Study Tips

1. **Same DNA, different cells** — Always remember this core concept; it explains why regulation matters
2. **Draw the levels** — Sketch a diagram from DNA → chromatin → transcription → mRNA → protein, and mark where regulation can occur
3. **Lac operon = model system** — Understand the logic: repressor blocks when substrate absent, releases when substrate present
4. **Acetylation = Activation** — Both start with "A"; acetylation loosens chromatin
5. **Think about calico cats** — A visual real-world example of X-inactivation and epigenetics

6. **Epigenetics $\neq$ mutation** — Epigenetic changes modify expression, not the DNA sequence itself